MUHAMMAD SHOAIB ASGHAR

Penetration Tester | Cybersecurity Analyst | SOC Analyst

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on cybersecurity experience gained through self-directed lab work: penetration testing (Nmap, Metasploit) against intentionally vulnerable systems, SIEM deployment and alert triage with Wazuh, and full-cycle OSINT investigations. Builds security tools independently, including an AI-assisted vulnerability scanner and a cryptographically sound password manager with a documented threat model. Proficient in Python, FastAPI, SQL, and Linux; working knowledge of MITRE ATT&CK, OWASP Top 10, and major compliance frameworks. Seeking an entry-level SOC or penetration testing role.

TECHNICAL SKILLS

Cybersecurity: SOC Operations · SIEM (Wazuh, Splunk) · Penetration Testing · Vulnerability Assessment · Threat Hunting · Threat Intelligence · Incident Response · Log Analysis · OSINT · Digital Forensics (Basic) · Risk Assessment · Security Monitoring · IOC Analysis

Security Tools: Nmap · Metasploit · Wireshark · Burp Suite · OWASP ZAP · Nessus · Snort · Kali Linux

Frameworks & Compliance: MITRE ATT&CK · OWASP Top 10 · ISO/IEC 27001 · GDPR · HIPAA · PCI DSS

Programming & Databases: Python · SQL · JavaScript · FastAPI · REST API · PostgreSQL · MySQL

Operating Systems: Linux · Windows

CYBERSECURITY PROJECTS

SILI — AI Cybersecurity Assistant

- Built a self-hosted AI security tool in Python that answers security questions via live web search and scans code for vulnerabilities.
- Integrated multiple AI APIs (Anthropic Claude, Groq, Serper, NVD) with a modular backend so the underlying model can be swapped without touching the rest of the app.
- Designed and built the entire application independently, including a custom terminal-style UI.

Secure Password Toolkit — Desktop Application

- Built a local-first credential manager in Python 3 (Tkinter) with authentication, encrypted storage, and a formal threat model and security assessment report.
- Implemented salted PBKDF2-HMAC-SHA256 key derivation (200,000 iterations) for login and per-entry encryption, keeping the session key in memory only.
- Enforced brute-force resistance (4-attempt lockout, 5-minute cooldown) and multi-user/multi-platform namespacing, with an optional Master Key for defense-in-depth.
- Added a real-time password strength meter and documented residual risks with hardening recommendations (e.g. migrating to AES-GCM).

Penetration Testing Lab — Metasploitable2

- Built a home penetration testing lab (Kali Linux + Metasploitable2) to practice the full attack lifecycle in a controlled environment.
- Performed reconnaissance with Nmap (service/version detection, default scripts) to enumerate open ports and running services (FTP, SSH, Telnet, SMB, RPC, web).
- Cross-referenced enumerated service versions against known vulnerabilities to prioritize exploitation targets.
- Used Metasploit to exploit vulnerable services, gain remote shell access, and perform post-exploitation checks (user/privilege enumeration).

OSINT Investigation Challenge — Security Blue Team

- Completed Security Blue Team's Introduction to OSINT challenge: a full open-source investigation from a single social media handle.

- Pivoted across Twitter/X, Blogger, and a self-owned website to build a subject profile, using CyberChef (Base64/Hex decoding) and Google dorking to uncover hidden accounts.
- Produced a structured write-up documenting methodology, evidence correlation (image/metadata matching), and findings.

Additional Projects

- **Wazuh SIEM Deployment and Log Analysis —** Wazuh SIEM Deployment: configured agents, analyzed logs, and generated alerts for real-time security monitoring.
- **Social Engineering Simulation Using SET —** Simulated phishing attacks and credential harvesting using the Social Engineer Toolkit (SET) in a controlled lab.
- **Cybersecurity Awareness Website with Chatbot —** Built a cybersecurity awareness website with an interactive chatbot covering phishing, malware, ransomware, and cyber hygiene.
- **Crime Investigation Management System —** Developed a secure REST API (FastAPI + PostgreSQL) with authentication, relational data modeling, validation, and AI-based risk scoring.

EXPERIENCE

App Developer — Digisol 365 | *Bahawalpur* | *Sep 2024 – Oct 2025*

- Managed IT operations and support, including system maintenance, troubleshooting, network management, and software solutions.
- Contributed to project management, team coordination, and client support, demonstrating strong technical expertise and leadership.

EDUCATION

BS Software Engineering — Islamia University Bahawalpur

CGPA: 3.34 / 4.00

- Advanced Python Bootcamp
- Web Development Bootcamp

CERTIFICATIONS

- Cyber Job Simulation — Deloitte, via Forage (Completed June 29, 2026; Enrolment Verification Code: C9Yo5RzfzJwoSY255)
- Introduction to OSINT — Centri / Security Blue Team (Certificate ID: 310180483, completed July 30, 2026)
- Introduction to Cybersecurity — Cisco (Credential ID: 29e9c9db-f072-43d1-8bc1-e476e4474bd5, issued July 22, 2026)

TRYHACKME — HANDS-ON PRACTICAL TRAINING

Profile: tryhackme.com/p/shoaibasgha

- Completed 26 rooms and earned 10 badges across OSINT (OhSINT), Windows exploitation (Blue — EternalBlue/MS17-010), Linux fundamentals, web security (OWASP Top 10), and SOC operations (closed first alert, 100% true positive rate in a scenario); ranked in the top 20% of users.
- Completed the free Jr Penetration Tester learning path, including Guided Pentest: Web and Guided Pentest: Infrastructure, practicing live attack-and-defense scenarios that reinforce reconnaissance, exploitation, and alert-triage skills used in the projects above.

SOFT SKILLS

Problem Solving · Critical Thinking · Team Collaboration · Communication · Technical Documentation · Time Management · Quick Learner · Adaptability